

Introduction

Sharps injuries occur frequently among healthcare workers and are often not reported. Therefore, it is vital that all members of staff minimize the risks of either themselves; their colleagues or their service users/relatives from receiving an accidental inoculation injury in the work place. At least 40% of the injuries could be prevented by the adoption of safer sharps handling practice. Recapping of used needles causes approximately one-third of injuries; sharps not discarded after use causes one-quarter of the injuries and a further 10% are associated with overfilled sharps bins.

The blood-borne viruses (BBV) Hepatitis B virus (HBV), Hepatitis C virus (HCV) and Human Immunodeficiency virus (HIV) can be acquired following occupational exposure to infected body fluids or blood. The estimated risk for transmission of blood borne viruses is as follows:

SEROCONVERSION RATE IF DONOR POSITIVE			
EXPOSURE	PERCUTANEOUS	MUCOUS MEMBRANE	BROKEN SKIN
HBV	30%	Less than 10%	Less than 10%
HCV	3%	Less than 5%	Less than 5%
HIV	0.3%	0.1%	<0.1%

Purpose:

- This policy outlines the management procedure of needle stick injuries and incidents involving exposure to blood and other body fluids.
- It also highlights the recommendations for the safe use and disposal of sharps in all areas within Royal Hospital.
- It provides risk assessment necessary to determine the need for Post Exposure Prophylaxis (PEP)
- It states the strategies required to reduce risk of exposures.
- It details the administrative responsibilities surrounding the management of exposures including notification procedures, the completion and submission of forms.

Scope

This policy applies to all staff employed or working for Royal Hospital such as students, residents, housekeeping staff who have sustained a needle stick injury or body fluid exposure incident during the course of their work, regardless of the location.

4. Definitions

An Exposure:

Is a percutaneous injury (e.g., a needle stick or cut with a sharp object) or contact of mucous membrane or non-intact skin (e.g., exposed skin that is chapped, abraded, or afflicted with dermatitis) with blood, tissue, or other body fluids that are potentially infectious.

Health-care Worker (HCW):

A person (e.g., employee, student, contractor, attending clinician, public-safety worker, or volunteer) whose activities involve contact with patients or with blood or other body fluids from patients in a health-care setting.

Blood borne viruses (BBV's):

Viruses, which are transmitted in the blood, may cause disease in other people. The most common BBV's are HBV, HCV and HIV.

Body fluid:

This refers to any human body fluid, which could act as a source of blood borne viruses. All exposures, whether through a penetrating injury or by splash onto eyes, mucosal surfaces or abrasions will be defined as a 'needle stick injury/body fluid exposure incident'.

Needle stick Injury (NSI)/ Inoculation Injury:

NSI is where a needle or sharp object, contaminated with blood or other body fluid, penetrates the skin. For the purpose of this policy, this also includes human bites and scratches that break the skin and mucosal surfaces, such as eyes or mouth.

Roles & Responsibilities**Individual Staff**

All staffs are responsible for:

- Demonstrating adherence to this policy at all times.
- Behaving in a safe and responsible manner taking all appropriate steps to minimize the risks of transmission of BBV's.
- Undertake Infection Prevention and Control training.
- In the event of sustaining a needle stick injury/body fluid incident follow the guidance outlined in the accidental inoculation injury algorithm.

Administration

They are responsible for ensuring availability of appropriate resources required for the implementation of this policy and ensure its effectiveness is continually reviewed.

Infection prevention & control department

- Educate HCWs on standard precautions, safe sharp handling and disposal
- Ensure all staff are aware of the actions and the management required following an inoculation injury and the urgency with which it should be dealt with.

- Ensure that the Hepatitis B vaccination is offered to any member of staff assessed on pre-employment to be at risk of exposure.
- Maintain the Hepatitis B vaccination Database.
- Keep records of inoculation injury incidents
- Follow up inoculation injury investigations including blood results of the source

Staff clinic

- Assess employees presenting following exposure incidents during office hours and manage them.
- Liaise with infectious disease (ID) Consultants, as appropriate for ongoing advice and follow-up.
- Ensure appropriate samples from employees are obtained

Adult emergency department:

- Assess exposed employees out of hours and other service users presenting following exposure incidents. They should be seen as high priority in A&E
- To commence PEP when clinically indicated following a risk assessment after consulting ID consultant.

Infectious Disease unit:

- Provide specialist advice on management of exposure incidents
- Undertake specialist counseling for HIV, HBV and HCV positive source patients.
- Undertake follow up for injured employees who are on PEP.

Microbiology laboratory:

- Prioritize testing of samples for the source and recipient on a timely manner
- Inform staff clinic of any positive results for BBV in recipient and/or source.

Line Managers

- Ensure that all staffs are familiar with the procedure outlined in this policy and the management of inoculation injury is embedded in their areas in accordance with policy and best practice.

Prevention of Needle stick Injuries and Blood Borne Virus Exposures

- Elimination - working practices should be regularly reviewed to wherever possible eliminate the use of unnecessary sharps.
- Education & training :
All staff must receive education and training about the prevention of needle stick injury and procedure to be followed should such incident occur at pre-employment and thereafter
- Engineering controls

Wherever possible medical devices incorporating safety protection mechanisms should be supplied to staff to use e.g. using a safety-Lock blood collection set in place of a needle and syringe.(currently not in place, a future plan)

- Safe systems of work

Managers will ensure safe systems of work are in place and staffs adhere to the waste management policy.

- PPE

Staff should use appropriate personal protective equipment such as gloves, visor, and apron for procedures where there is a risk of blood or body fluid exposure.

- Vaccination

Staff that is at risk of exposure to blood & body fluids must be vaccinated against HBV

Promotion of safe sharps practice and the prevention of sharps injury:

STANDARD PRECAUTIONS

All blood, tissues and body fluids, should be regarded as potentially infectious

- Wear gloves where contact with blood or body fluids is anticipated. If there are cuts/abrasions on the hands, these should already be covered in an appropriate waterproof dressing and gloves worn before contact with blood/body fluids.
- Take extreme care when handling and disposing of used sharps. Carry the sharp box at the point of care.
- Do not hand over the sharp to another person. Dispose of in the sharp box immediately
- Discard needle and syringe as one unit, whenever possible.
- Label sharps bins with the date assembled, date closed, name of source department and hospital.
- Carry sharps bins by the handle and away from your body.
- Do NOT recap used needles. If recapping is unavoidable use a one-handed technique.
- Do NOT pass used sharps from person to person by hand, e.g. from surgeon to scrub nurse – use a receiver or similar container.

- Do not place sharps bins within reach of un-authorized persons, especially children. They must be placed above waist level in an appropriate area.
- Do NOT overfill sharps bins. When $\frac{3}{4}$ full sharps bins must be properly closed and sealed. It is everyone's responsibility to replace sharps bins when $\frac{3}{4}$ full.
- Do not place sharps bins in a yellow clinical waste bag.

Prevention of contamination of skin lesions

- Cuts, abrasions, chapping and eczema on the hands present a hazard and must be covered with a waterproof dressing or gloves.
- Anyone likely to come into contact with blood or other body fluids of any patient must wear disposable gloves.
- The use of gloves does not protect from inoculation injury but should heighten the awareness that a high-risk procedure is being carried out. Gloves will protect cuts and lesions on the hands from contact with blood and body fluids.
- It is strongly recommended that individuals with eczema/psoriasis report to staff clinic if the condition exacerbates. Splashes of blood/body fluids onto normal, i.e. undamaged skin, does not pose a risk of transmission of blood-borne viruses.

Preventing inoculation injuries during surgical procedures

- Have no more than one person working in an open wound/body cavity at any time (unless essential to the safe and successful outcome of an operation).
- Use a 'hands-free' technique where the same sharp instrument is not touched by more than one person at the same time; avoid hand to hand passing of sharp instruments during an operation.
- Assure safer passage of necessary sharp needles and instruments via a 'neutral zone'
- Announce when a sharp instrument or needle is placed there. The 'neutral zone' may be suitable tray, kidney basin or an identified area in the operative field.
- Ensure that scalpels and sharps needles are not left exposed in the operative field, but always removed promptly, appropriately and safely by the scrub nurse from the 'neutral zone'.

- Use instruments rather than fingers for retraction, and for holding tissues while suturing
- Direct sharp suture needles away from own non-dominant, or assistant's hand.
- Remove sharp suture needles before tying suture; tie suture with instruments rather than with fingers. Consider double gloving with a larger pair of gloves innermost for optimum comfort. Double gloving won't prevent a sharps injury but will reduce the volume of blood transmitted to the HCW.
- If a glove puncture is suspected or recognized, rescrub if possible and re-glove as soon as safety permits.
- Change gloves regularly if performing, or assisting with a prolonged surgical procedure even if no glove puncture is suspected or recognized.
- Protect mucous membranes of eyes with protective eyewear where there is risk of blood/body fluid splashing. This should prevent splash injuries (including lateral splashes).
- Choose waterproof gowns, or wear a surgical gown with waterproof cuffs and sleeves and a plastic apron underneath for procedures likely to cause a high blood loss.
- Feet should be protected by using boots or clogs.
- Wear protective headwear and surgical mask appropriately.
- Ensure that all blood is thoroughly removed / cleansed from a patient's skin at the end of an operation before patient leaves theatre.
- Remove all protective clothing including footwear on leaving the contaminated area.

Courses of action after exposure to blood/body fluid:

Refer to Appendix A: Accidental Inoculation Injury (AI) algorithm

Immediate First Aid

This action to be done by the victim who sustained injury or exposed to blood and or body fluids.

Percutaneous injuries:

- Wash the injury site under running water, but not by sucking or squeezing. There is no documented evidence to support that squeezing the wound will further reduce the risk of transmission of blood borne infection.
- Then wash the wound with soap and water without scrubbing for at least five minutes.
- Antiseptics, bleach and skin washes should not be used as there is no evidence of their efficacy to reduce the risk of infection.
- Dry and cover the wound with a waterproof plaster.

Splashes onto non-intact skin (e.g. abrasions, cuts, and eczema):

- Wash liberally with water.

Splashes into the eyes, mouth or other mucous membranes:

- Irrigate with copious amounts of clean water. If wearing contact lenses, leave them in place while irrigating, as they form a barrier over the eye and will help protect it. Once the eye has been cleaned, remove the contact lenses and clean them in the normal manner. This will make them safe to wear again.

Reporting

- The exposure incident must be reported immediately to the appropriate administrative personnel, usually the immediate supervisor of the unit/ward where the exposure occurred.
- Complete Accidental Inoculation Injury Reporting Form (**Appendix B**), including the name and hospital number of the source patient, if known.
- Report the incident to staff clinic during working hours. Outside normal working hours, staff must report the injury to the nurse in charge of the clinical area or their supervisor/manager and then report to the Adult Emergency Department.
- HCWs should report to infection control department for risk assessment of exposure, counseling procedure and follow up of results

- If there is a high risk of HIV transmission, post-exposure prophylaxis (PEP) needs to be commenced ideally within an hour of injury.
- The on-call for Infectious Diseases can be contacted for advice on risk assessment, counseling and need for PEP and must be contacted if the risk is high or involves a known positive patient.
- The Line Manager/Supervisor should ensure completion of Accidental Inoculation Injury Reporting Form.
- The ICP should organize blood tests for hepatitis B surface antigen, hepatitis C antibody and HIV antibody from source patient during working hours. The manager in the concerned area should organize testing of the source patient outside working hours.
- The source patient should be approached by one of the members of the clinical team caring for the patient, or the doctor/nurse manager who is on-call for that patient out of hours.

Assessing the Risk of Infection after Exposure

The following factors are associated with an increased risk of BBV transmission:

- Deep injury.
- Hollow needle.
- Visible blood on device that caused injury.
- Injury with a needle that had been placed in a source patient's artery or vein.
- The risk of Hepatitis- B transmission is increased if the source patient is HBeAg positive.
- A high HIV plasma viral load in the source is associated with an increased risk of HIV transmission.

To assess the risk of infection after exposure and decide the significance of exposure, refer to tables 1-3 and the flowcharts in appendix C1 & C 2.

Management of accidental exposure to HBV:

1. Establish whether a significant injury has occurred
2. Check the immunity status of the exposed HCW and follow **table 4** for management.

For significant injuries:

Consider the need for Hepatitis- B vaccine +/- Hepatitis B immunoglobulin (HBIG) using table 4. Where indicated, treatment should be given as soon as possible after injury. In most circumstances, treatment can be initiated without waiting for source patient blood test results. Ideally, both HBV vaccination and HBIG should be given

within 48 hours of injury, although they should still be considered up to a week after exposure

11. Management of HCV exposure

There is no vaccine or post exposure prophylaxis presently available for HCV. There are effective treatments however, and it is therefore important that those exposed receive appropriate follow-up to enable treatment initiation should they become infected.

If source is known hepatitis C positive take baseline HCV serology. Follow up blood test should be done as outlined in table 6 in appendix D

Management of HIV exposure

Clinical evaluation & baseline testing of the exposed HCW:

HCW exposed to HIV should be evaluated within hours (rather than days) after their exposure and should be tested for HIV at baseline.

Timing & duration of PEP for HIV

For purposes of considering HIV PEP, the evaluation also should include information about medications the exposed person might be taking and any current or underlying medical conditions or circumstances (i.e., pregnancy, breast feeding, or renal or hepatic disease) that might influence drug selection.

If a high-risk exposure has occurred, PEP should be commenced. Ideally the first dose should be given within one hour of the incident. However the exact mechanisms of HIV infection are not fully understood, therefore if more than one hour has elapsed after exposure this does not automatically preclude starting PEP. PEP should be started even when the interval since exposure exceeds 36 hours. Initiating therapy after a longer interval (e.g. 1 week) might be considered for exposures that represent an increased risk for transmission.

Use of PEP When HIV Infection Status of Source Person is Unknown:

If the source person's HIV infection status is unknown at the time of exposure, use of PEP should be decided on a case-by-case basis, after considering the type of exposure and the clinical and/or epidemiologic likelihood of HIV infection in the source. If these considerations suggest a possibility for HIV transmission and HIV testing of the source person is pending, initiate PEP regimen until laboratory results have been obtained and later modify or discontinue the regimen accordingly.

The following are recommendations regarding HIV post-exposure prophylaxis:

- If indicated, start PEP as soon as possible after an exposure.

- Re-evaluation of the exposed person should be considered within 72 hours post-exposure, especially as additional information about the exposure or source person becomes available.
- Administer PEP for 4 weeks, if tolerated.
- If a source person is determined to be HIV-negative, PEP should be discontinued.
- Follow **Table 6:** Recommended HIV post exposure prophylaxis for percutaneous injuries. For PEP regimen, please refer the exposed person to infectious disease specialist for advice.

Follow-up of HCW Exposed to HIV:

HCW with occupational exposure to HIV should receive follow up counselling, post-exposure testing, and medical evaluation, regardless of whether they receive PEP. Follow **table 7:** Follow up Testing of HCWs exposed to HIV.

Discontinuing Post-exposure Prophylaxis

If the Exposed has begun chemoprophylaxis and the Source subsequently tests negative, chemoprophylaxis should be discontinued. In these situations, it should be emphasized to the exposed that there is no benefit to completing the medication(s) in the starter kit. Continuation of chemoprophylaxis might be considered in rare instances where there is concern about the source being in the window period of infection (sero-conversion phase).

Review and monitoring

- This policy is reviewed every three years and monitored for effectiveness by annual audit of the Needle stick Incident Database (NSI) to review compliance with the process for the management of an inoculation incident.
- The results will be reported to infection prevention & control where any recommendations and actions for improvement are made.
- Quarterly incident statistics on sharps injury is reported to the central infection prevention & control at MoH.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Integrated Post-exposure Protocol for HIV, HBV and HCV: Guidelines for Managing Exposures to Blood and Body Fluids.2009.		2009	
Policy for the Prevention and Management of Needle stick Injuries and Blood Borne Virus Exposures.2013.		2013	
Peninsula Community Health. Needle stick		2011	

Injury Policy - The Management of Body Fluid Exposure Incidents including needle stick, scratches, bites and other bodily fluid exposure. 2011			
National Health Safety. Management of Needle stick Injuries and Incidents Involving Exposure to Blood and Body Fluids.2013.		2013	
National Health Safety. Management of Needle stick Injuries and Incidents Involving Exposure to Blood and Body Fluids.2011.		2011	

Review / Revision History

Revision History			
Revision	Description of change	Author	Revision Date
1	Initial Release	Aisha Al Mamari, Dr Amal Al Tai, Dr Zaina Al Maskari	6/10/22
2			
3			
4			
Written by		Checked by	Authorized by
Aisha Al Mamari Dr Zaina AL Maskari Dr Amal AL Tai		Dr Omaira Al Omairi	Dr Amal Al Tai

Appendices:

Appendix A: Accidental Inoculation Injury (All) Algorithm

Appendix B: Accidental Blood and Body fluids Exposure Reporting Form

Appendix C1: Assessment of risk of injury

Appendix C2: Risk assessment and management of HCWs exposed to BBV

Tables:

Table 1: Assessment of risk of injury type

Table 2: Assessment of risk of body fluid

Table 3: Assessment of the exposure source

Table 4: Management of Exposure to HBV positive source

Table 5: Follow up testing for the victim post exposure to HCV positive patient

Table 6: Recommended HIV post exposure prophylaxis for percutaneous injuries

Table 7: Follow up Testing of HCWs exposed to HIV

Appendix A: Accidental inoculation injury (All) Algorithm

DEFINITION

- Percutaneous injury: any injury with contaminated sharp object
- Mucocutaneous: exposure of a mucous membrane (eye, nose, mouth or non-intact skin with blood or body fluid)

Action after All

◆ After percutaneous injury:

- Encourage the site to bleed, ideally by holding it under running water(don't suck or squeeze).
- Wash the wound or exposed area with soap and water
- Dry the injured site and cover with a waterproof plaster or dressing

◆ After mucocutaneous exposure:

- Splashes on eyes should be irrigated with sterile water for 10 minutes
- HCWs wearing contact lenses should irrigate before and after removal of lenses
- Splashes on the skin should be irrigated with copious amount of water for 10 minutes

- **Inform immediate supervisor in clinical area**
- **Complete the All incident reporting form**
- **Submit the form to Infection prevention & control (IP&C)**

During working hours:

Report to staff clinic for assessment and investigations.
Submit the form to IP&C.

Outside working hours:

Report to Accident & Emergency for assessment and investigations.
Submit the form to IP&C next working day.

Victim blood investigations

HBsAg, Anti-HBc Total ,Anti HBs titer (if no previous records) , HCV & HIV 1&2 serology.

Source blood investigations:

HBsAg, Anti-HCV, HIV 1&2 serology
Check previous records before sending
Make sure source blood sample is collected and sent to the lab.

Follow up requirement and treatment advice will be decided after risk assessment (see algorithms 1 & 2)

Appendix B: Accidental Blood and Body fluids Exposure Reporting Form

	سلطنة عُمان وزارة الصحة SULTANATE OF OMAN MINISTRY OF HEALTH	Directorate General for Disease Surveillance & Control Central Department of Infection Prevention & Control Hospital/Health Institution:								
ACCIDENTAL BLOOD & BODY FLUIDS EXPOSURE REPORTING FORM										
For Immediate In-Charge Use										
Name	Patient ID	Ward/ Clinic	Designation	Staff Number	Date of Injury	Time of Injury				
Exposed Person										
Source										
Hepatitis Vaccine History for exposed person		1st Dose / /	2nd Dose / /	3rd Dose / /	Level Of Anti-Hbs IU/mL					
Percutaneous Injury (Needle Stick / Sharp Injury)				Mucotaneous Exposure (Blood Splash or Body Fluids Splash)						
Used syringe needle	Un-used Needle	Surgical Knife	Others	Material	Mouth	Eye	Nose	Intact skin	Non intact skin	Others
Site of Injury		Severity		Estimated Amount and Type of Blood/ Body Fluid						
First Aid	Percutaneous Injury		Done	Not Done	Mucotaneous Exposure		Done	Not Done		
	Allow the site to bleed				Remove contaminated clothing					
	Wash with soap and water				Irrigate affected area with copious amount of water / normal saline/ soap					
	Cover with appropriate bandages									
Please Describe Briefly and Concisely How The Accident Occurred										
.....										
Exposed Person Signature:										
For Treating Physician Use										
Risk Assessment										
Urgency of Testing	Risk Factors of Source					Yes	No	Comments		
Most Urgent (testing within next 24 hours)	Consider high risk if met one of the following: • Known case HBV, HCV, HIV • Immigration from an endemic country • History of IV drug user • High risk sexual activity • History of acute liver disease • History of Blood product / Organ recipient									
Urgent Testing (within 36-48 hours)	When the exposure is highly significant and source status is unknown									
NOT Very Urgent (first next working day)	When the exposure is low significant and source status is known as negative									

For Infection Prevention and Control Use			
Date of Injury	/	/	Date of Reporting
			/
			/

Lab Investigation Result						
Test	Test Markers	Source	Exposed Person			
		Day of Exposure	Day of Exposure	After Six Weeks	After Three Month	After Six Month
Hepatitis B	HBsAg		/	/	/	/
	Anti-HBs					
Hepatitis C *(if source HCV positive)	Anti-HCV					
HIV	Anti-HIV					

Treatment Started			
Hepatitis B immunoglobulin (If source is HBV+ and exposed person not immunized/ non-responder)	Date of Receiving	Dose Given	
	/		
Date of Hepatitis B vaccine	1st dose	2nd dose	3rd dose
Post vaccination Anti-HBs level	Date of Test	< 10 IU/mL	> 10 IU/mL
	/	IU/mL	IU/mL
Anti-retro Viral Medication (If source known to HIV)	Date of Receiving	Name of Medication	Dose Started
	/	1.....	1.....
	Date of Stopping	2.....	2.....
	/	3.....	3.....

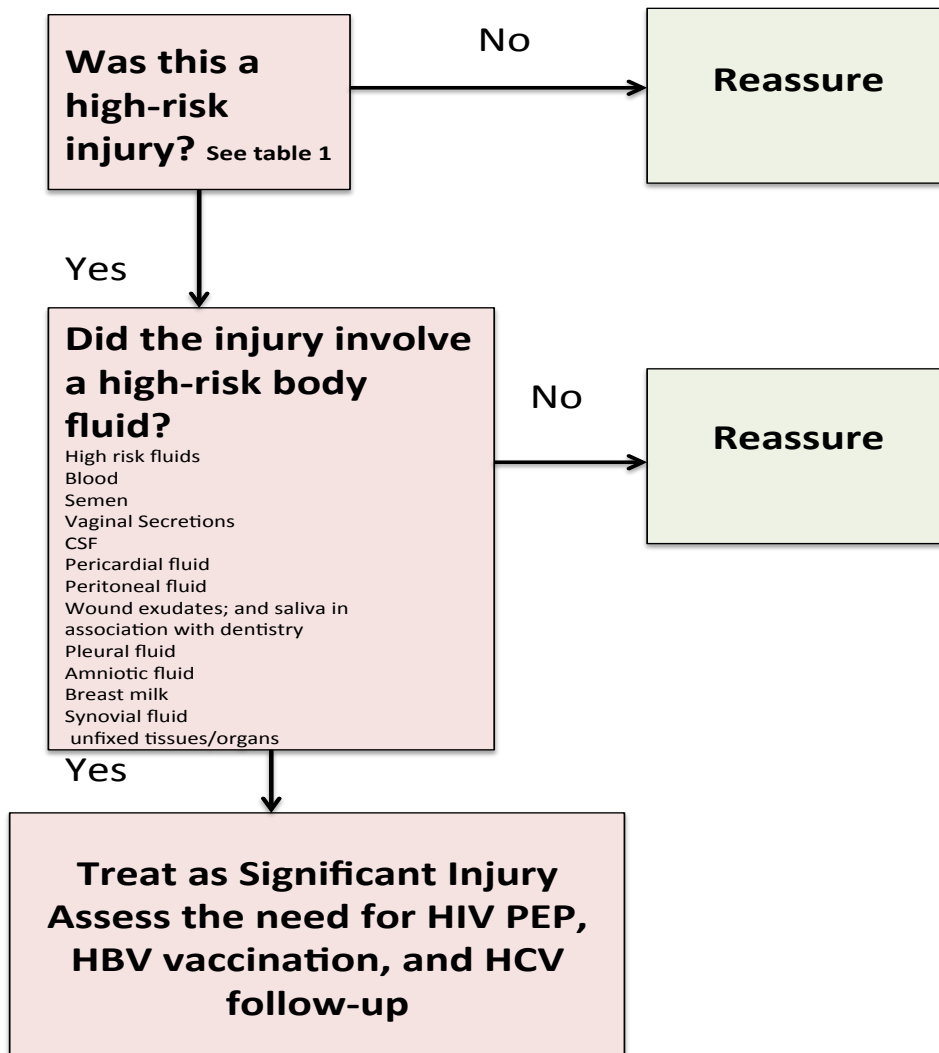
Health Education		
Counseling to exposed person to known or yet to be identified to HBV, HCV or/and HIV source, for the period of follow up (six months):	Explained	Not Explained
1. Refrain from donating blood, semen, plasma or tissue		
2. Pregnant or lactating women should be advised against breast feeding		
3. Personal items such as toothbrushes and razors should not be shared		
4. Sexual intercourse should be protected		

Infection Prevention and Control Interpretation	
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Infection Prevention and Control Name :	Signature :

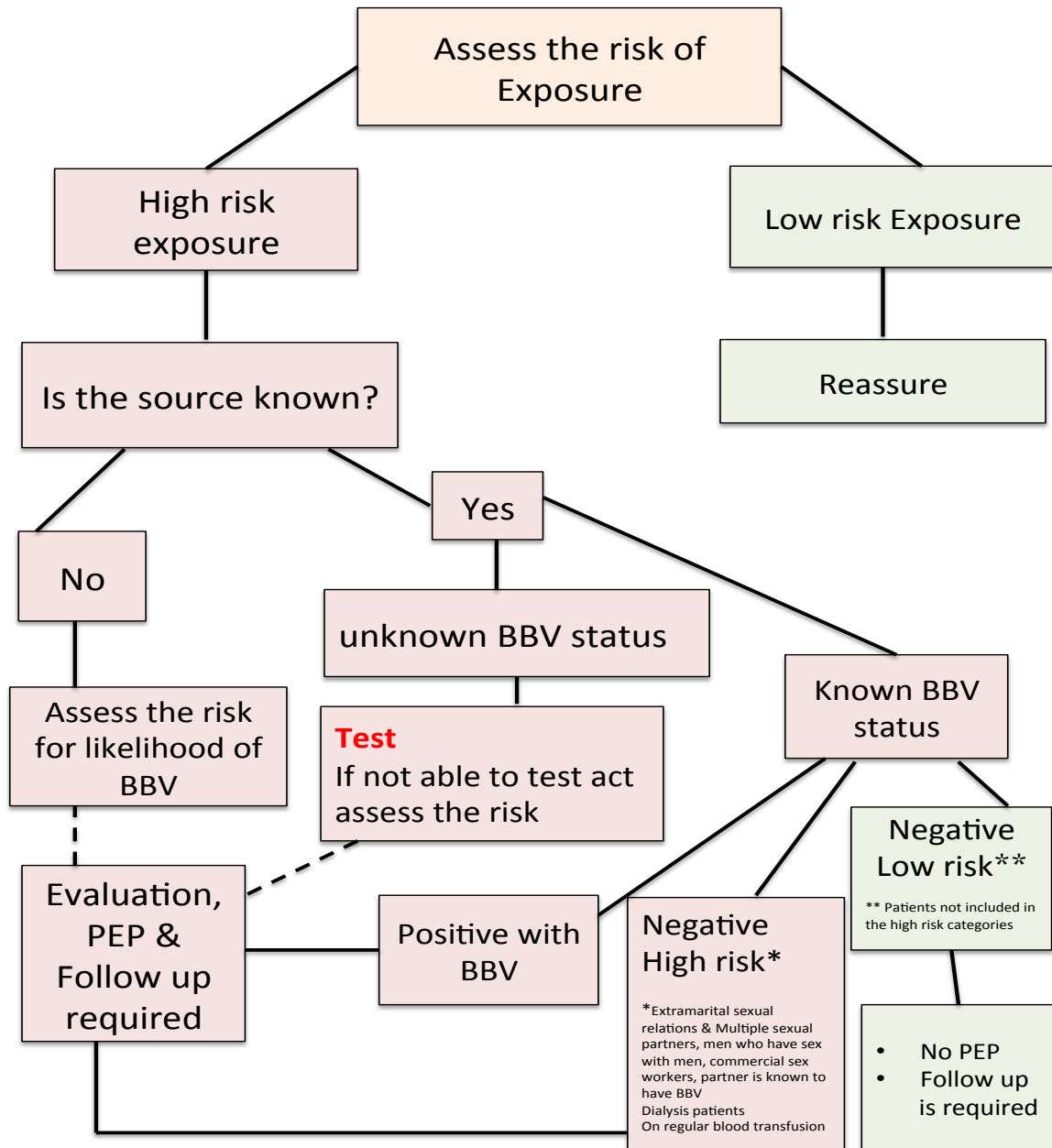
- Note:**
1. Post vaccination screening for Anti-HBs is required at 1-2 months following completion of the vaccine.
 2. Personnel who do not respond to the primary vaccine series should be revaccinated and tested to Anti-HBs 1-2 month following the last vaccination.
 3. If the following completion of the 2 vaccine series the Anti-HBs level is <10 IU/mL, the employee should be counseled about the result and the importance of standard precautions.
 4. If source known to HIV, anti-retro viral medication to be started as soon as possible with in 24hrs for exposed person and not later than 72 hours.
 5. If source known to HBV, HBIG to be given as soon as possible (within 48hrs and no later than 7 days) to exposed person if not vaccinated or not responding to vaccine or revaccination schedules.
 6. If source known to HCV, no recommendation for the use of IG or interferon, follow-up LFT for exposed person.

Appendix C1 : Assessment of risk of injury

For an injury to be considered **significant** in terms of BBV transmission, **both** the type of injury incurred **and** the body fluid involved must be **high-risk**.



Appendix C2: Risk assessment and management of HCWs exposed to BBV



Tables (1-3) showing the risk categorization related to the NSI:

Table 1: Risk from Injury Type

High-Risk Injury – ‘SIGNIFICANT’	Low-Risk Injury
Percutaneous exposure e.g. needle stick or other sharps injury Exposure on any broken skin Mucous membrane exposure (e.g. eye) Human bites	Splash on intact skin – there is no known risk of BBV transmission from exposures to intact skin.

Table 2: Risk of Body Fluid Type

High-Risk Body Fluid – ‘SIGNIFICANT’	Low-Risk Body Fluid (unless blood-stained)
Blood Semen Vaginal Secretions CSF Pericardial fluid Peritoneal fluid Wound exudates; and saliva in association with dentistry Pleural fluid Amniotic fluid Breast milk Synovial fluid Unfixed tissues or organs	Urine Sputum Vomit Faeces Saliva Tears

Table 3: Assessment of the exposure source

Status of the source	Actions
Source is known	Check laboratory records for recent results if unknown status test for BBV HBsAg Anti-HCV Ab Anti-HIV Ag/Ab
Source is known but unable to test	Assess for likelihood of BBV risk factors: Extramarital sexual relations & Multiple sexual partners, men who have sex with men, commercial sex workers, partner is known to have BBV, known IV drug abuser, or on hemodialysis
Source is unknown	Evaluate the likelihood of exposure to a source at high risk for infection

Table 4: Management of Exposure to HBV positive source

HBV status of person exposed	HbsAg positive source	Unknown source	HbsAg negative source
Known responder Of HB vaccine (anti-HBs >10mIU/ml)	- Consider booster dose of HB vaccine - No follow up testing required	- Consider booster dose of HB vaccine - No follow up testing required	No treatment or follow up required
> 2 Doses of HB vaccine given, or course completed but response unknown.	- One dose of HB vaccine followed by second dose one month later - Check anti-HBsAb Level after 3 month of vaccine	- One dose of HB vaccine followed by second dose one month later - Check anti-HBsAb Level after 3 month of vaccine	- One dose of HB vaccine followed by second dose one month later - Check anti-HBsAb Level after 3 month of vaccine
Unvaccinated or Previously had 1dose HB vaccine pre-exposure	- Accelerated course of HB vaccine* plus HBIG x 1 - Check anti HBsAb level after 1 month - If anti HBsAb is negative check Anti-HBcAb and HBsAg, at 3 months and 6 months from exposure	- Accelerated course of HB vaccine* plus HBIG x 1 - Check anti HBsAb level after 1 month. - If anti HBsAb is negative check Anti-HBcAb and HBsAg, at 3 months and 6 months from exposure	- Initiate or complete course of HB vaccine - Check anti-HBsAb Level after 3 month of vaccine
Known non responder to HB vaccine (anti- HBs<10mIU/ml) 2-4 months post immunization	- HBIG x 1 - Consider booster dose of HB vaccine. - A second dose Of HBIG should be given at one month - Check Anti-HBcAb & HBsAg at 3 and 6 months from exposure - If HCW did not receive second series of HB vaccine or anti HBsAb titer is unknown	- HBIG x 1 - Consider booster dose of HB vaccine. - A second dose Of HBIG should be given at one month	No treatment or follow up

- Persons who have previously been infected with HBV are immune to reinfection and do not require post exposure prophylaxis.
- Hepatitis B immune globulin; dose is 0.06 mL/kg intramuscularly.
- A responder is a person with adequate levels of serum antibody to HBsAg (i.e., anti-HBs >10 mIU/mL).
- A nonresponder is a person with inadequate response to vaccination (i.e., serum anti-HBs < 10 mIU/mL).
- An accelerated course of vaccine consists of doses spaced at zero, one and two months. A booster dose may be given at 12 months to those at continuing risk of exposure to HBV.

Table.5 Follow up testing for the victim post exposure to HCV positive patient

Blood for HCV status	6 weeks	3 months	6 months
HCV antibody test	No	Yes	Yes
HCV RNA (PCR)	Yes	Yes	No

Table 6: Recommended HIV post exposure prophylaxis for percutaneous injuries

Exposure type	Infection status of source			
	HIV Positive	Source of unknown HIV status †	Unknown source	HIV negative
Less severe ¶	Recommend PEP	Generally, no PEP warranted; however, consider PEP* for source with HIV risk factors ††	Generally, no PEP warranted; however, consider PEP* in settings where exposure to HIV infected persons is likely	No PEP warranted
More severe	Recommend PEP	Generally, no PEP warranted; however, Consider PEP* for source with HIV risk factors ††	Generally, no PEP warranted; however, consider PEP* in setting where exposure to HIV-infected persons is likely	No PEP warranted

† Source of unknown HIV status (e.g., deceased source person with no samples available for HIV testing).

§ Unknown source (e.g., a needle from a sharps disposal container).

¶ Less severe (e.g., solid needle and superficial injury).

§§ More severe (e.g., large-bore hollow needle, deep puncture, visible blood on device, or needle used in patient's artery or vein).

*The designation "consider PEP" indicates that PEP is optional and should be based on an individualized decision between the exposed person and the treating clinician.

†† If PEP is offered and taken and the source is later determined to be HIV-negative, PEP should be discontinued.

Table 7: Follow up Testing of HCWs exposed to HIV:

Blood for HIV status	6 weeks	3 months	6 months
HIV antibody test	Yes	Yes	Yes
- Extended HIV follow-up (e.g., for 12 months) is recommended for HCW who become infected with HCV following exposure to a source co-infected with HIV and HCV. - HIV testing should be performed on any exposed person who has an illness that is compatible with an acute retroviral syndrome, regardless of the interval since exposure.			